

L2. Photons and Their Quantum States in a Complex Vector Space

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From Numbers to Complex Vectors

- When we speak of computing, we think about numbers.
 - The set of natural numbers $\{1, 2, 3, \dots\}$ is denoted by $\mathbb{N}$.
 - The set of integers $\{\dots, -2, -1, 0, 1, 2, \dots\}$ is denoted by $\mathbb{Z}$.
 - The set of rational numbers in the form of $r = p/q$, where $p, q \in \mathbb{Z}$ and $q \neq 0$, is denoted by $\mathbb{Q}$.
 - The set of real numbers is denoted by $\mathbb{R}$. In particular, probabilities are real numbers whose sum is normalized to 1.
 - The set of complex numbers is denoted by $\mathbb{C}$.
- Observe that

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}.$$

- The values of physical attributes of classical systems are mostly real numbers. In high dimensions, we encounter real vector spaces, e.g., the velocity of an air molecule.
- Sometimes, for computational convenience, we introduce complex numbers in classical electrodynamics. For instance, we unify R, L, C in an AC circuit as a collection of complex impedances and solve with complex amplitudes of current and voltage.
- Quantum mechanics, on the other hand, is founded intrinsically on complex vector spaces. Immediate questions are: Of what is the vector space? Why complex? How to generalize probability to quantify quantum information?

Outline

- Photon Polarization, an Example
- Dirac's Bracket Notation
 - Vector Space and Vectors
 - Complex Conjugate and Inner Product
 - Orthonormal Basis and Completeness
 - Projection Operators
- Photon Polarization in the New Language
- Lab 2 Application: Quantum Key Distribution

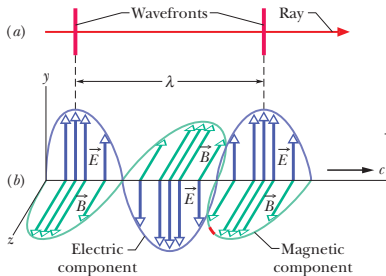
2.1 Photon Polarization, an Example

Take-Home Messages

- The polarization of a photon is an example of a two-dimensional complex vector space (i.e., a qubit). we can use a complex vector to represent an arbitrary polarization.
- The polarization of a photon can be measured by an analyzer, which allows the photon to pass with a probability.

The Polarization of Light

- The polarization of a photon is our first physical example of a vector space (which can, naturally, be used as a qubit). Before we dive into the quantum description later, let us review the classical understandings of light.
- **(Property I)** Light is an electromagnetic wave. It is transverse, which means that the electric (and magnetic) field of a light wave is perpendicular to the propagation direction.



- Recall that a planar and monochromatic scalar wave traveling in the z direction has an amplitude of vibration

$$u(z, t) = u_0 \cos(kz - \omega t + \delta),$$

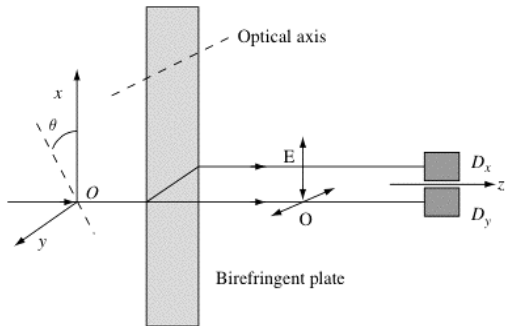
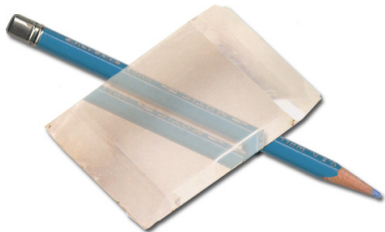
where the speed of the wave $c = \omega/k$.

- At $z = 0$ (which can be generalized for arbitrary z),

$$u(t) \equiv u(z = 0, t) = u_0 \cos(\omega t - \delta).$$

- For an electromagnetic wave, what oscillates is an E vector in the x - y plane (we ignore B for simplicity).

- **(Property II)** Light waves (with vector nature of $\vec{E}$) can be decomposed into two rays by birefringent crystals,¹ such as Iceland spar.



¹Crystals with non-cubic crystal structures are often birefringent, as are plastics under mechanical stress.

- For light with linear polarization,

$$E_x = E_0 \cos \theta \cos \omega t,$$

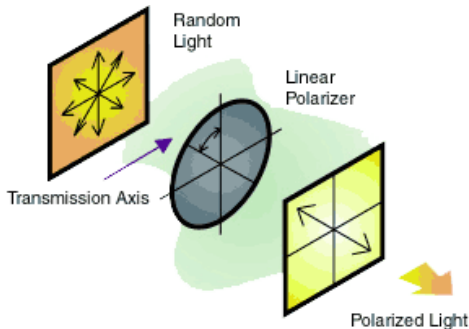
$$E_y = E_0 \sin \theta \cos \omega t.$$

So we can write

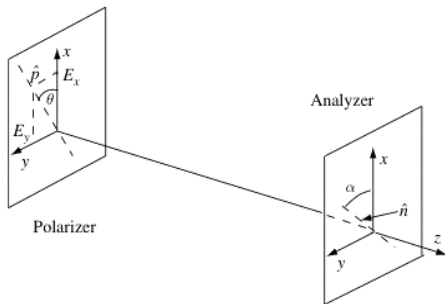
$$\vec{E} = E_0 \hat{p} \cos \omega t, \quad \hat{p} = (\cos \theta, \sin \theta).$$

- The intensity of the light is proportional to the square electric field $I \propto E_0^2$.

- The linear polarization can be achieved by letting random light pass through a polarizer whose axis makes an angle θ with x axis.



- We then pass the polarized light through an analyzer, whose axis is along $\hat{n}$.



- At the exit from the analyzer the electric field $\vec{E}'$ is obtained by projecting the field $\vec{E}$ onto $\hat{n}$

$$\vec{E}' = (\vec{E} \cdot \hat{n})\hat{n} = [E_0(\hat{p} \cdot \hat{n})]\hat{n} \cos \omega t$$

- We obtain the **Malus law** $I' = I(\hat{p} \cdot \hat{n})^2$.

The Quantum Interpretation

- Light is composed of photons, or light quanta (Einstein).
 - A photon can be emitted from an atom, when an electron jump from higher to lower energy levels.
 - A photon can be detected by a photodetector (or CCD camera).
- The Malus law must have a statistical interpretation, as photons cannot be further divided.
 - Out of N photons with the polarization vector $\hat{p} = (\cos \theta, \sin \theta)$, approximately (within statistical fluctuations $\sim \sqrt{N}$)

$$\frac{N_{\hat{n}}}{N} \simeq (\hat{p} \cdot \hat{n})^2$$

photons pass through the analyzer with axis along $\hat{n}$.

Elliptical Polarization

- Imagine we use a birefringent crystal to split the horizontal and vertical components of a linearly polarized light, and introduce phase delays δ_x and δ_y , respectively.
- As a result, we obtain elliptical polarization

$$E_x = E_0 \cos \theta \cos(\omega t - \delta_x) = E_0 \operatorname{Re}(\mu e^{-i\omega t}),$$
$$E_y = E_0 \sin \theta \cos(\omega t - \delta_y) = E_0 \operatorname{Re}(\nu e^{-i\omega t}),$$

where $\mu = \cos \theta e^{i\delta_x}$ and $\nu = \sin \theta e^{i\delta_y}$, so $|\mu|^2 + |\nu|^2 = 1$.

- It is important to note that only $\delta = \delta_y - \delta_x$ is physically relevant. This is because, by a time shift, we can choose $\delta_x = 0$.

- Therefore, we can use a complex vector $(\mu, \nu)^T$, where $|\mu|^2 + |\nu|^2 = 1$, to represent an arbitrary polarization.
 - The polarization vector is the same as $(\mu e^{i\phi}, \nu e^{i\phi})^T$ for arbitrary ϕ , so there are only two real parameters.
- The need to use a complex representation is not just a mathematical convenience. It comes from the fact that $\cos \omega t$ and $\sin \omega t$ are linearly independent and span an extra two-dimensional space.
- In a pedagogical way, we have just introduced a complex vector space with a probability interpretation for the mathematical description of photon polarization.

- In this course, we will pay more attention to information, probability, observables, and their relationship, not to matter, energy, waves or particles.
- Therefore, instead of dealing with historical events, differential equations, and the almost incomprehensible wave-particle duality (as we did in General Physics), we introduce quantum mechanics through the generalization of the complex vector space with its probabilistic meaning.

2.2 Dirac's Bracket Notation

Take-Home Messages

- A complex vector space is composed of elements $|a\rangle$ called **kets**. They are column vectors in our usual notation. It has a dual space whose elements are **bras** $\langle a|$.
- **Inner product** $\langle a|b\rangle$ generalizes the dot product of usual vectors. Notice $\langle a|b\rangle = \langle b|a\rangle^*$.
- Choosing an **orthonormal basis** of kets $|i\rangle$ (notice $\langle i|j\rangle = \delta_{ij}$), we can expand an arbitrary ket $|a\rangle$ as

$$|a\rangle = \sum_i |i\rangle \langle i|a\rangle.$$

Vector Space in Quantum Mechanics

- In quantum mechanics, a vector space is composed of elements $|a\rangle$, known as **kets** or ket-vectors (following Dirac's invention). It satisfies the following axioms.

- 1 The sum of any two kets is also a ket:

$$|a\rangle + |b\rangle = |c\rangle.$$

- 2 Vector addition is commutative:

$$|a\rangle + |b\rangle = |b\rangle + |a\rangle.$$

- 3 Vector addition is associative:

$$(|a\rangle + |b\rangle) + |c\rangle = |a\rangle + (|b\rangle + |c\rangle).$$

- (Continued)

- ④ There is a unique vector 0 such that when you add it to any ket, it gives the same ket back:

$$|a\rangle + 0 = |a\rangle.$$

- ⑤ Given any ket $|a\rangle$, there is a unique $-|a\rangle$ such that

$$|a\rangle + (-|a\rangle) = 0.$$

- Up to this point, ordinary vectors also satisfy the same set of axioms. (We now define the complex, as oppose to real, vector space).

- The following two axioms involves the multiplication by any complex number. Ordinary vectors, however, can only be multiplied by real numbers.
 - ⑥ Given any ket $|a\rangle$ and any complex number ζ , you can multiply them to get a new ket. Also multiplication by a scalar is linear:

$$|a'\rangle \equiv |\zeta a\rangle = \zeta|a\rangle.$$

- ⑦ The distributive property holds:

$$\zeta(|a\rangle + |b\rangle) = \zeta|a\rangle + \zeta|b\rangle,$$

$$(\zeta + \omega)|a\rangle = \zeta|a\rangle + \omega|a\rangle.$$

- Concrete examples of complex vector spaces include:
 - The set of two-dimensional column vectors with complex elements, in the form

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

- The set of continuous complex-valued functions $\psi(x, y, z, t)$ of spacetime variables x, y, z, t .
- They (separately or combined) are the most frequently used representations of quantum states (or wave functions) in this course.

The Complex Conjugate Vector Space

- A complex vector space has a dual space, i.e., the complex conjugate vector space.
- For every ket $|a\rangle$, there is a **bra** or bra-vector $\langle a|$ in the dual space. Bras satisfy the same axiom as kets, e.g.,
 - Suppose $\langle a|$ and $\langle b|$ are the bras corresponding to the kets $|a\rangle$ and $|b\rangle$, respectively. Then the bra corresponding to $|a\rangle + |b\rangle$ is $\langle a| + \langle b|$.
- It is worth noting:
 - The bra corresponding to $|\zeta a\rangle$ for any complex number ζ is

$$\langle \zeta a| = \langle a|\zeta^* = \zeta^* \langle a|.$$

Length and Angle from Inner Product

- The analog to the dot product for ordinary vectors is the **inner product** for a bra and a ket is $\langle b|a\rangle$, which is a complex number.
 - The inner products are linear:

$$\langle c|(|a\rangle + |b\rangle) = \langle c|a\rangle + \langle c|b\rangle,$$

$$(\langle a| + \langle b|)|c\rangle = \langle a|c\rangle + \langle b|c\rangle.$$

- Interchanging bras and kets corresponds to complex conjugation:

$$\langle b|a\rangle = \langle a|b\rangle^*.$$

Hence, $\langle a|a\rangle$ is a real number.

- For example, if ket $|a\rangle$ and $|b\rangle$ are represented by the column vectors

$$|a\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \end{pmatrix}, \quad |b\rangle = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \end{pmatrix},$$

then the corresponding bra $\langle b|$ is represented by the row vector

$$\langle b| = (\beta_1^* \quad \beta_2^* \quad \beta_3^* \quad \beta_4^* \quad \beta_5^*).$$

- The inner product is defined as the product of the row vector and column vector:

$$\begin{aligned}\langle b|a\rangle &= (\beta_1^* \quad \beta_2^* \quad \beta_3^* \quad \beta_4^* \quad \beta_5^*) \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \\ \alpha_5 \end{pmatrix} \\ &= \beta_1^* \alpha_1 + \beta_2^* \alpha_2 + \beta_3^* \alpha_3 + \beta_4^* \alpha_4 + \beta_5^* \alpha_5.\end{aligned}$$

- Observe that $\langle b|a\rangle = \langle a|b\rangle^*$.

Normalized and Orthogonal Vectors

- Normalized vector is a generalization of unit vector in ordinary space. Normalized vectors satisfy

$$\langle a|a\rangle = 1.$$

- In ordinary space, vectors are orthogonal if their dot product is zero (they are also said to be perpendicular to each other). In Hilbert space, two vectors are orthogonal if

$$\langle b|a\rangle = 0.$$

Basis and Basis Vectors

- Let us consider a set of n vectors $\{|x_1\rangle, \dots, |x_n\rangle\}$ in $V = \mathbb{C}^n$. The set is said to be **linearly dependent** if the equation

$$\sum_{i=1}^n \eta_i |x_i\rangle = |0\rangle$$

has a solution $\eta_1, \dots, \eta_n$, at least one of which is non-vanishing.

- If the trivial solution $\eta_i = 0$ ($1 \leq i \leq n$) is the only solution, the set is said to be **linearly independent**.

- The set of n linearly independent vectors is called a **basis** of $\mathbb{C}^n$ and the vectors are called **basis vectors**.
- For example, a set of vectors

$$v_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

is a basis of $\mathbb{C}^2$. However, the set of basis vectors are not mutually orthogonal. Neither are they normalized.

Orthonormal Basis and Completeness

- In ordinary space, it is useful to define a set of mutually orthogonal unit vectors and use them as a basis to construct vector, e.g.,

$$\vec{v} = x\hat{i} + y\hat{j} + z\hat{k}.$$

- In a complex vector space, we can also find a set of orthonormal basis of kets labeled by $|i\rangle$, where $i = 1, 2, \dots, N$. The dimension of the Hilbert space is the maximum number of mutually orthogonal vectors.

- Any vector $|a\rangle$ can be written as a sum of basis

$$|a\rangle = \sum_i \alpha_i |i\rangle,$$

where the complex number α_i are called the components of the vector.

- We can calculate an individual component corresponding to the basis vector $|j\rangle$ as

$$\alpha_j = \langle j|a\rangle,$$

since $\langle j|i\rangle = \delta_{ij}$ (i.e., 1 if $i = j$ and 0 otherwise).

- Therefore, we can also rewrite the decomposition of the vector $|a\rangle$ in a more elegant form as

$$|a\rangle = \sum_i |i\rangle \langle i|a\rangle,$$

which implies the following identity:

$$\sum_i |i\rangle \langle i| = I,$$

where I is an identity operator (note that we will come back to the definition of operator).

Projection Operators (Polarizers/Analyzers)

- The matrix

$$P_k \equiv |k\rangle\langle k|$$

introduced above is called a **projection operator** in the direction defined by $|k\rangle$. This projects a vector $|v\rangle$ to a vector parallel to $|k\rangle$ in such a way that $|v\rangle - P_k|v\rangle$ is orthogonal to $|k\rangle$.

- The set $\{P_k = |k\rangle\langle k|\}$ satisfies the conditions
 - ❶ $P_k^2 = P_k$,
 - ❷ $P_k P_j = 0$ ($k \neq j$),
 - ❸ $\sum_k P_k = I$ (completeness relation).

2.3 Photon Polarization in the New Language

- Using the mathematical language of vector space, we can revisit the polarization of photons and write:
 - Photon polarization lives in a 2D complex vector space. An orthonormal basis can be chosen to be linear polarization along x and y axes.

$$|h\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |v\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

- The corresponding projection operators (or filters) are

$$P_h = |h\rangle\langle h| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_v = |v\rangle\langle v| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

We confirm the completeness condition $P_h + P_v = I_{2 \times 2}$.

- A general state $|p\rangle$ of a photon is represented by a unit vector

$$|p\rangle = I_{2 \times 2}|p\rangle = |h\rangle\langle h|p\rangle + |v\rangle\langle v|p\rangle = \mu|h\rangle + \nu|v\rangle,$$

where $\mu, \nu \in \mathbb{C}$ and $\langle p|p\rangle = |\mu|^2 + |\nu|^2 = 1$.

- The components of $|p\rangle$ (i.e., a generalization of $\hat{p} \cdot \hat{n}$),

$$\mu = \langle h|p\rangle, \quad \nu = \langle v|p\rangle$$

are known as the **probability amplitudes**.

- The probabilities for the measurements of horizontal and vertical polarizations are given by

$$P_h = |\mu|^2 = \langle p|h\rangle\langle h|p\rangle \equiv \langle p|P_h|p\rangle,$$

$$P_v = |\nu|^2 = \langle p|v\rangle\langle v|p\rangle \equiv \langle p|P_v|p\rangle.$$

- The total probability $P_h + P_v = 1$.
- After the measurement, the photon is either in the state $|h\rangle$ or in the state $|v\rangle$. (Strictly speaking, this depends on the detail of the measurement; photon can be destroyed. At this stage, we do not worry about it.)

Discussion

- We consider the following four types of linearly polarized photons with polarization vectors $\vec{e} = (\cos \zeta, \sin \zeta)$:

ζ	Arrow	Representation	Mode
0	$\longleftrightarrow$	$ h\rangle$	I
$\pi/2$	$\updownarrow$	$ v\rangle$	I
$\pi/4$	$\nearrow$	$(h\rangle + v\rangle)/\sqrt{2}$	II
$3\pi/4$	$\nwarrow$	$(- h\rangle + v\rangle)/\sqrt{2}$	II

- What is the probability of detecting $\longleftrightarrow$ for a photon in $\nwarrow$?
- What is the probability of detecting $\nearrow$ for a photon in $\longleftrightarrow$?

Summary

- We made the mathematical preparations for some general principles of quantum mechanics.
 - The state of a quantum system is represented by a unit vector in a vector space of states.
 - The squared magnitudes of the components of the state-vector, along particular basis vectors, represents probabilities for the corresponding measurements.
- Pedagogically, we introduced our first quantum information system—photons. We can encode information with their polarization. In the homework, we will use them to illustrate the interference phenomenon, a manifestation of quantum strangeness.

- Dirac's notation is so clear and efficient that it practically rules the quantum mechanical world. (Here, L represents a linear operator, which we formally introduce in Lecture 3.)
 - Everything in the form of $\langle \dots \rangle$ is a number, such as

$$\langle b|a\rangle, \quad \langle b|L|a\rangle, \quad \dots$$

- Everything in the form of $|\dots\rangle$ or $\langle\dots|$ is a vector, such as

$$|a\rangle, \quad L|a\rangle, \quad \langle b|, \quad \langle b|L, \quad \dots$$

- Finally, $|a\rangle\langle b|$ is an operator, also known as the **outer product**, as oppose to the inner product $\langle b|a\rangle$. In particular, assuming that $|\psi\rangle$ is normalized, $|\psi\rangle\langle\psi|$ is a projector.

References

- L. Susskind and A. Friedman, Quantum Mechanics (Basic Books, 2014), Chaps. 1-2.
- M. Le Bellac, A Short Introduction to Quantum Information and Quantum Computation (Cambridge, 2006), Chap. 2.
- M. Nakahara and T. Ohmi, Quantum Computing: From Linear Algebra to Physical Realizations (CRC Press, 2008), Chap. 1.

P2-1. Circular polarization is defined by

$$E_x = \frac{E_0}{\sqrt{2}} \cos \omega t,$$

$$E_y = \frac{E_0}{\sqrt{2}} \cos \left(\omega t \pm \frac{\pi}{2} \right) = \mp \frac{E_0}{\sqrt{2}} \sin \omega t,$$

where the phase shifts $\pm\pi/2$ correspond to the left- and right-handed polarized photons. The tip of the electric field vector traces a circle clockwise or counter-clockwise, respectively.

As in the lecture, an orthonormal basis for photons can be chosen to be linear polarization along x and y axes.

$$|h\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |v\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Justify the following expressions for the right- and left-handed polarized photons:

$$|R\rangle = \frac{1}{\sqrt{2}}(|h\rangle + i|v\rangle), \quad |L\rangle = \frac{1}{\sqrt{2}}(|h\rangle - i|v\rangle).$$

P2-2. Using the basis $|h\rangle$ and $|v\rangle$, we can write down

$$|+\rangle = \frac{1}{\sqrt{2}}(|h\rangle + |v\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(-|h\rangle + |v\rangle),$$

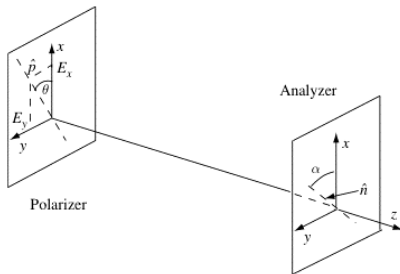
which represent photons linearly polarized along directions making an angle $\pm\pi/4$ with the x axis, respectively. Define

$$|R'\rangle = \frac{1}{\sqrt{2}}(|+\rangle + i|-\rangle), \quad |L'\rangle = \frac{1}{\sqrt{2}}(|+\rangle - i|-\rangle).$$

How are $|R'\rangle$ and $|L'\rangle$ related to $|R\rangle$ and $|L\rangle$? Do these state vectors represent physical states different from $|R\rangle$ and $|L\rangle$? If not, then why not?

P2-3. Classical versus Quantum Randomness

- Assume a polarizer has a polarization axis $\hat{p} = (\cos \theta, \sin \theta)$. An analyzer, on the other hand, has a polarization axis $\hat{n} = (\cos \alpha, \sin \alpha)$.



- A photon passing through the polarizer has a state, in Dirac's notation,

$$|p\rangle = \cos \theta |h\rangle + \sin \theta |v\rangle.$$

- Between the polarizer and analyzer the photon can be either in the horizontal polarization $|h\rangle$ (with probability amplitude $\langle h|p\rangle = \cos \theta$) or in the vertical polarization $|v\rangle$ (with probability amplitude $\langle v|p\rangle = \sin \theta$).
 - For $|h\rangle$ photon, the probability amplitude of being measured by the analyzer is $\langle n|h\rangle = \cos \alpha$.
 - For $|v\rangle$ photon, the probability amplitude is $\langle n|v\rangle = \sin \alpha$.
- Therefore, there are two indistinguishable paths for measuring a photon polarized along $|n\rangle$, $p \rightarrow h \rightarrow n$ and $p \rightarrow v \rightarrow n$.

- (a) Show that, if we follow the classical probability theory, i.e., by tossing a coin at the polarizer/analyzer and adding probabilities of the two paths, we expect to observe with a total probability

$$P_{\text{tot}}^{\text{classical}} = |\langle p|h\rangle\langle h|n\rangle|^2 + |\langle p|v\rangle\langle v|n\rangle|^2.$$

- (b) In the current setup, the Malus law becomes $I' = I \cos^2(\theta - \alpha)$. Show that the law implies

$$P_{\text{tot}}^{\text{quantum}} = |\langle n|h\rangle\langle h|p\rangle + \langle n|v\rangle\langle v|p\rangle|^2.$$

Compare the results to the phenomenon of interference.

Lab 2 Application: Quantum Key Distribution

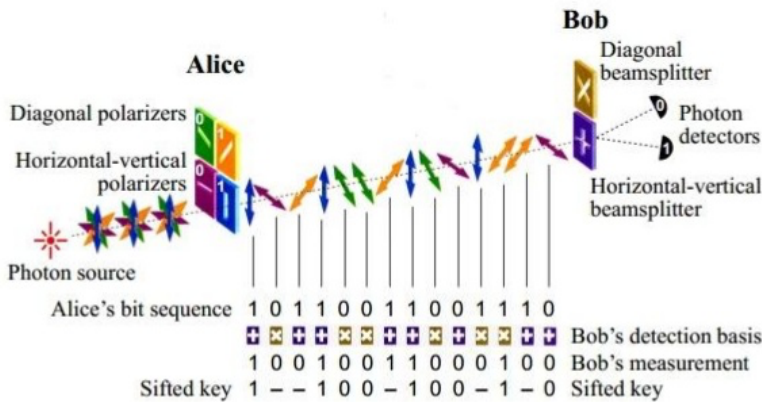
- Goal: To send encrypted messages that cannot be understood by anyone but the designated recipient.
- Message is a whole number m , e.g., represented by the dots and dashes of Morse code as ones and zeros.
- An encryption is a function $f: m \longrightarrow f(m)$, agreed on between Alice (sender) and Bob (recipient) but unknown to Eve (a possible eavesdropper).
- Problem: If the same encryption is used many times, Eve can usually deduce the nature of the encryption and read the messages (by, e.g., frequency analysis).

- Classical solution: Let the encryption depend on a frequently changed key, which can be regarded as another whole number k . The encrypted message is now $f(m, k)$. For example, $f(m, k) = km$. (In this case, the time Eve needs to factorize the product km grows faster than any power of km .)
- New problem: Alice and Bob must frequently exchange messages to establish new keys, and these new messages too may be intercepted by Eve.
- Quantum solution: *It is not possible to measure any quantity without changing an unknown state vector to one in which that quantity has some definite value.*

The BB84 Protocol

- Alice sends the key to Bob as a sequence of linearly polarized photons with polarization vectors of the form $\vec{e} = (\cos \zeta, \sin \zeta)$, where ζ are various angles.
- Alice represents ones and zeros by values of ζ in either one of the two modes.

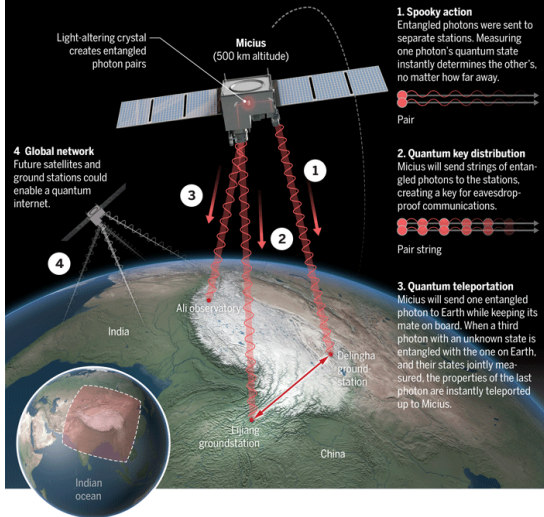
Mode	Binary digit	ζ	Arrow
I	0	0	$\longleftrightarrow$
	1	$\pi/2$	$\updownarrow$
II	0	$\pi/4$	$\nearrow$
	1	$3\pi/4$	$\nwarrow$



- Finally, Alice and Bob communicate over a classical, possibly public channel to compare their choices of basis for each bit. The bits for which Alice and Bob have used different bases are discarded.

Quantum leaps

China's Micius satellite, launched in August 2016, has now validated across a record 1200 kilometers the "spooky action" that Albert Einstein abhorred (1). The team is planning other quantum tricks (2-4).



[doi:10.1126/science.aan6972]

"It's a huge, major achievement," says Thomas Jennewein, a physicist at the University of Waterloo in Canada. "They started with this bold idea and managed to do it."

S.-K. Liao et al.,
Satellite-to-ground quantum key distribution,
Nature 549, 43 (2017).